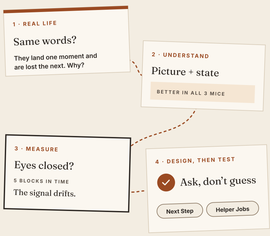
An ASDRP neuroscience portfolio · Amishi Mehta

From neurons to next steps



At Pragnya's outings for neurodivergent kids and adults, and in everyday schoolwork, the same instruction can land one moment and be lost the next. That raised a question: **does what a brain is doing (its state: alert, tired, calm) change what it takes in?** We tested it in two public brain recordings, tried hard to break each answer, and turned what held up into the design of two apps.

Read the story → | Start with the mouse study

The same story as slides, a narrated video and a phone tour. Public data only. A prediction is not a cause.

Understand
Mouse: state changes how the brain sees

Measure
EEG: readable once, then it drifts

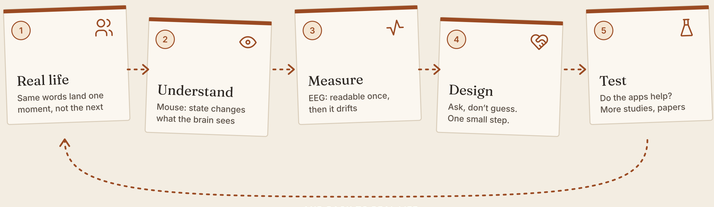
Checked
Every result tested to break it

Design
Findings become app features

Chapter 5: where this goes next, with more studies, design research and papers.

One loop, five chapters

It starts in real life and ends back there. Each chapter answers the question the last one raised.



1 · REAL LIFE

Same words, different moment.

At Pragnya's outings and in schoolwork, a step that is easy when someone is calm can be lost when they are tired, rushed or overloaded. Is that the brain itself?

To find out: look inside a brain while its state changes.

2 · UNDERSTAND

A mouse's state changes how its brain sees.

Mice watched pictures while thousands of brain cells were recorded. We guessed what each cell would do from the picture alone, then from the picture plus how alert and active the mouse was.

Found: knowing the mouse's state made our guesses 25–33% better, in all three mice. Read chapter 2 →

3 · MEASURE

Can sensors read state in a person?

A cap of 14 sensors on one person's head for two minutes (EEG). Closing your eyes is one of the clearest changes it picks up. Can a simple model tell when?

Found: on one stretch, often (0.625, where 0.5 is a coin flip). Across the whole recording it drifted to 0.478. Brain-wave bands did better (0.566) but still dipped below guessing. Read chapter 3 →

4 · DESIGN

From findings to features.

If state matters but cannot be read from outside, an app should ask, not guess. It should show less on a low-energy day, and make pausing and coming back easy.

Built into: Next Step, by Amishi Mehta, for schoolwork, and Helper Jobs, a family project, built for real jobs at Pragnya's Sunday outings. Read chapter 4 →

5 · TEST, THEN GO FURTHER

Where this goes next.

One follow-up done, three more data experiments, design studies with Pragnya volunteers and families, and papers others can check.

The aim: one research direction, where neuroscience and design meet. Read chapter 5 →

The loop: real life raised the question; the mouse showed that state changes what a brain takes in; the EEG showed that state is hard to read from outside; so the apps ask how someone is doing and keep each step small. The last experiment takes the apps back to real life to see if they help.
Three mice and one person are a start, not a rule for everyone.

What this does not show yet: whether the pattern holds in more animals and more people, what causes it, or whether the apps help. Each of those is an experiment in chapter 5.

How we tried to break each result

Nothing in the story counts until it has survived a check that could have made it disappear.



MIX UP THE ANSWERS

It should fail, and it does

If we scramble which moments were eyes-closed, the model drops to a coin flip. If we scramble the mouse's state, the gain disappears. So the results come from real signal.

NEVER PEEK AT THE FUTURE

Learn from earlier, test on later

The model only ever learns from the past and is tested on what comes after, the way a real app would have to work.

LEAVE A GAP

No help from next-door moments

Moments right next to each other look alike. We leave a gap between what the model learns from and what it is tested on.

LEAVE ONE MOUSE OUT

No single mouse carries it

Take any one mouse away and the other two still show the gain.

LOOK FOR BROKEN READINGS

Four found and capped

Four readings were up to 180 times bigger than normal. Capping them helps one test, but does not fix the drift.

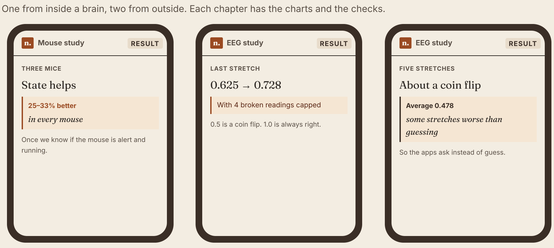
EVERY NUMBER TRACED

Rebuilt from saved results

Every chart is drawn from the saved results files, and the EEG study reruns with one command.

The story in three numbers

One from inside a brain, two from outside. Each chapter has the charts and the checks.



State matters inside

Knowing the mouse's state made its brain a quarter to a third easier to predict.

Readable once...

On the last stretch, a simple model could often tell eyes open from closed.

...but not over time

Within two minutes, the brain-wave pattern for eyes closed changed.

Who did what
The studies were done by Amishi with her father, who helped shape the questions, review the code and fix problems. Software tools, including AI assistants, helped build this website and the apps. The contribution ledger lists each part.

What it is not
It is not a medical study and it does not diagnose anyone. Three mice and one person cannot speak for everyone, and being able to predict something is not the same as knowing what causes it.

Data: Sensorium 2022 (Willeke et al., 2022) and UCI EEG Eye State (Roesler, 2013). Mouse study · EEG study · Design · What's next

← All chapters

CHAPTER 2 · UNDERSTAND · MOUSE STUDY

Does a mouse's state change how its brain responds to what it sees?

Mice watched pictures while scientists recorded thousands of brain cells in the part of the brain that handles seeing. We tried to guess what each cell would do, first from the picture alone, then from the picture plus two signs of the mouse's state: how wide its pupils were and how fast it was running. If state did not matter, the extra signs would not help.



It mattered, in all three mice. Knowing the mouse's state made our guesses a quarter to a third better (25–33%), on pictures the model had never seen.
This shows state and seeing are linked. It does not yet show which causes which.



How we tested it

Two guessers, the same pictures, and a fair test on pictures neither had seen.

STEP 1

What the mouse saw

Each picture is shrunk to a short summary of its light and dark patterns, so a simple model can use it.

STEP 2

What state it was in

Pupil size (wider usually means more alert) and running speed, recorded at the same moment as the brain cells.

STEP 3

Two simple guessers

Guesser A sees only the picture. Guesser B sees the picture plus pupil and running. Both are the simplest kind of model: each clue gets a weight, and the weights are added up.

STEP 4

A fair test

Both guess how each brain cell responds to pictures they have never seen. We score how closely each guess follows the real cell.

THE DATA

A public dataset

Sensorium 2022: recordings of thousands of brain cells from mice watching pictures, shared by scientists for anyone to study.

BEING FAIR

Each mouse counts once

The many brain cells in one mouse are not counted as many mice.

The results

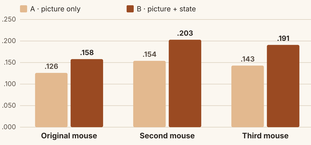


HOW TO READ THE SCORE

How closely the guess follows the real brain cell.

0 means the guess and the cell have nothing in common; 1 would be a perfect match. Brain cells are noisy, so even good guesses score low. What matters is the jump from A to B.

The score is a correlation, averaged over every recorded brain cell.

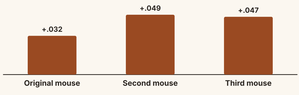


RESULT

Every mouse is easier to predict once we know its state.

The picture explains part of what the seeing part of the brain does. Whether the mouse is alert and moving explains more, in every animal.

Tested on pictures the guessers never saw.



HOW MUCH STATE ADDS

Every mouse gains, by different amounts.

Pupil size helps all three mice. How much running adds changes from mouse to mouse, so each mouse is shown on its own.

Gain = guesser B's score minus guesser A's score.

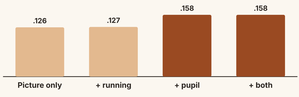
Inside the first mouse

From Amishi's saved results and report in Amishi123/neuroscience-project, redrawn here. These are for the first mouse only.

90%
of brain cells were guessed better once state was added

+0.017
typical gain per cell (the middle value)

+0.35
the biggest gain for a single cell

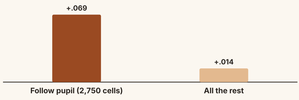


WHICH PART OF STATE?

It is the pupil, not the running.

Adding running barely moved the score. Adding pupil size gave almost the whole gain. This mouse ran on only about 5% of trials, so running had little to say.

Score on unseen pictures, first mouse.

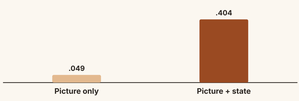


WHICH CELLS GAIN?

The cells that follow alertness gain most.

Cells whose activity rises or falls with pupil size gained about five times as much as the rest.

A cell "follows pupil" when its activity and pupil size move together (correlation above 0.1), as in Amishi's report.



ONE CELL

From noise to a pattern.

Brain cell 3035 was almost impossible to predict from the picture alone. Knowing the mouse's state raised its score from 0.05 to 0.40.

Score for this one cell on unseen pictures.

Trying to break the result

Three checks that could have made the gain disappear. It survived all three.

MIX UP THE STATE

The gain disappears

We gave guesser B the pupil and running from the wrong moments, 20 times per mouse. It added nothing. The gain needs the mouse's state at that moment.

LEAVE ONE MOUSE OUT

Still there

Take any one mouse away and the other two still show the gain, so no single mouse carries the result.

PUPIL OR RUNNING?

Pupil helps every mouse

Taking each sign away in turn: pupil helps all three mice; running helps some more than others.



LEAVE ONE MOUSE OUT

The gain stays without any one mouse.

So the pattern does not hang on one recording. It is a check, not a promise about all mice.

Average gain over the two mice that are left.

Code and data

The analysis is in Amishi123/neuroscience-project (her notebooks 01–05). The numbers on this page are copied from its saved results, and an automatic check confirms they still match. Data: Willeke et al. (2022), *The Sensorium competition on predicting large-scale mouse primary visual cortex activity*, arXiv:2206.08666.

What it does not show

Three mice cannot speak for all mice. A better guess does not show that being alert or running causes the brain to change. Other things that happen at the same time could explain it.

What chapter 2 tells us: the same picture lands differently depending on the state the mouse is in. So the next question is whether that state can be read from outside the head, in a person. Chapter 3 →

← All chapters Chapter 3 · EEG →

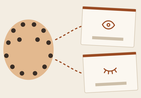
Everything in this portfolio: The website | Phone tour | Slides | Video | Next Step app | Helper Jobs app | Source code

← All chapters

CHAPTER 3 · MEASURE · EEG STUDY

Can sensors on the head tell when someone's eyes are closed?

EEG is a cap of small sensors that pick up the brain's electrical activity through the scalp. Closing your eyes is one of the clearest changes it can see. If a simple model cannot read even that reliably for two minutes, quieter states like tiredness will be harder still.



On one stretch, yes. Over time, no. On the last part of the recording the model scored 0.625, and 0.728 once four broken readings were capped. But tested on five stretches across the recording, it averaged 0.478: about a coin flip.
Letting it see the last few moments as well did not help.

How we tested it

Learn from earlier in the recording, then test on what comes later, like a real app would have to.



THE RECORDING

One person, two minutes

14 sensors, about 128 readings a second. Each moment is marked eyes open or eyes closed (closed about 45% of the time).

LEARN, THEN TEST

Earlier, then later

The model learns from the first 70% of the recording, is checked on the next 15%, and is tested once on the last 15%.

TWO MODELS

Now, or now plus a moment ago

One sees a single moment. The other also sees the last 9 readings, about a fourteenth of a second.

THE SCORE

0.5 is a coin flip

How often the model ranks an eyes-closed moment above an eyes-open one. 0.5 is guessing, 1.0 is always right.

FIVE STRETCHES

Moving forward in time

Learn from everything before a stretch, test on that stretch, and repeat five times across the recording.

THE DATA

A public recording

UCI EEG Eye State (Roesler, 2013), shared for anyone to study.

The results

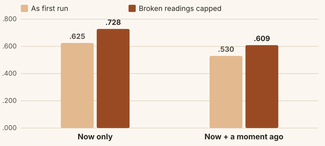


HOW TO READ THE SCORE

Where the results sit between guessing and always right.

Anything near 0.5 is no better than flipping a coin. Below 0.5 means the model is wrong more often than right, so the pattern it learned has flipped.

The score is called AUC.

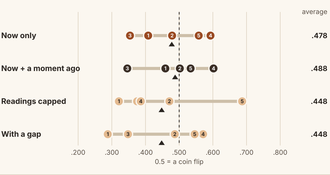


THE LAST STRETCH

Good on one stretch, better once broken readings are capped.

Four readings were broken: up to 715,897 when normal is about 4,000. Capping each sensor at the range seen while learning lifts the score from 0.625 to 0.728.

We added this check after the first results, so the new numbers sit next to the old ones, not in place of them.



FIVE STRETCHES IN TIME

Across the recording, it drifts to a coin flip.

Scores land on both sides of 0.5, and some fall well below it. Being worse than guessing means the pattern the model learned earlier had flipped by later on.

So the signal drifts within two minutes. Capping broken readings and leaving a gap do not fix it.



Making sure it is not our mistake

Three checks that the result comes from the recording, not from how we handled it.

MIX UP THE ANSWERS

Scores 0.499, a coin flip

With eyes open and closed scrambled 20 times, the model scores a coin flip, as it should. Our method does not invent a pattern.

LEAVE A GAP

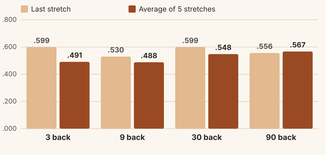
Scores 0.448, still near guessing

We left out the moments just before each test stretch, so the model could not lean on near-copies. The result stays near a coin flip.

CAP BROKEN READINGS

Four readings, capped fairly

The caps come only from the part the model learns from, so the test part never shapes them.



HOW FAR BACK TO LOOK

Looking further back seems to help, but we did not pick a winner.

We chose to try 3, 9, 30 and 90 readings back before seeing any results. Ninety does best on average, but picking it now would mean tuning to this one recording.

Next: compare these on new recordings before choosing.

Experiment 2: would brain-wave bands hold steady?

Alpha waves grow when eyes close. So we asked a follow-up question, wrote the plan down before running it, and tested it the same way.



THE IDEA

Strength over the last second

Instead of the raw reading at one moment, the model sees how strong four kinds of brain wave were over the last second, including alpha.

THE TEST

Same five stretches, plus a gap

Learn from earlier, test on later, with one second left out between them, because the one-second windows overlap.

THE RULE SET BEFOREHAND

Better on average, never below 0.5

It counts as holding up only if it beats the raw readings on average and no stretch falls below a coin flip.

Better on average, but it still did not hold up. All four bands averaged **0.566** across the five stretches, against **0.481** for the raw readings. But two stretches still dipped just below a coin flip (0.492 and 0.493), so by the rule we set beforehand it did not hold up. Alpha alone did worse (0.442).
Mixing up the answers gave 0.500, as it should.



EXPERIMENT 2

Brain-wave bands help, but not enough to rely on.

Bands lift the average and the worst stretch, but a model that is sometimes worse than guessing is still not something to build an app on. This is why the apps ask the person.

With a one-second gap between learning and testing. Without the gap: 0.571 for bands, 0.529 for raw readings. Plan written before running: EXPERIMENT-2-PLAN.md in the source code.

Rerun it
`python3 projects/eeg-eye-state/src/analyze.py` writes `results/metrics.json`, `src/analyze_bands.py` writes `results/bands.json` for experiment 2, and `python3 website/build_results.py` turns it into these charts. See the full report. Data: Roesler, O. (2013). *EEG Eye State*. UCI Machine Learning Repository, doi:10.24432/C57G7J.

What it does not show
One person and one short recording cannot show how this works for other people. The only thing measured is eyes open or closed, not attention, tiredness or any condition.

What chapter 3 tells us: a sensor that is right on one stretch and wrong on the next, even using brain-wave bands, is not something to build a tool on. So the apps ask the person instead of guessing. Chapter 4 →

← Chapter 2 · Mouse | Chapter 4 · Design → | All chapters

Everything in this portfolio: The website | Phone tour | Slides | Video | Next Step app | Helper Jobs app | Source code

← All chapters

CHAPTER 4 · DESIGN · HUMAN–COMPUTER INTERACTION

From findings to features.

Chapter 2 showed that a brain's state changes what gets through. Chapter 3 showed that state is hard to read from outside. Each finding became a design rule, and each rule is a real feature in two working apps.



Four findings, four design rules

Read each card top to bottom: what we found, the rule it gives, and where the rule lives.

FROM CHAPTER 2

Show less when energy is low.

Found: the same picture lands differently depending on the brain's state.

In the apps: Next Step asks "How much energy do you have today?" On "A little", it shows just the next step. Helper Jobs shows one picture step at a time, read aloud.

FROM CHAPTER 3

Ask, don't guess.

Found: a sensor read eye state on one stretch, then drifted to a coin flip. Brain-wave bands did not fix it.

In the apps: the person tells the app how they are doing. The energy check is optional, Help is a button they press, and nothing happens until they tap.

FROM CHAPTER 3

Expect change halfway through.

Found: within two minutes, the link between signal and state shifted.

In the apps: Next Step has Save & pause, a note of where you stopped, and "Pick up where you left off". Helper Jobs reminds helpers to take breaks.

FROM REAL LIFE

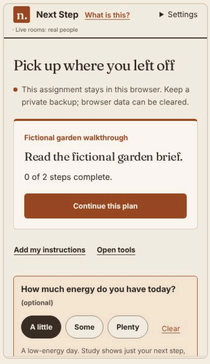
When stuck, make it smaller.

Seen: a task that is too big all at once is where people stop.

In the apps: Next Step's "I'm stuck" offers to make the step smaller, ask a question or take a break. Nothing is sent, and your place is kept.

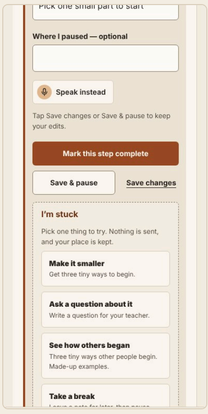
The rules, on real screens

Screens from the working apps, with made-up examples.



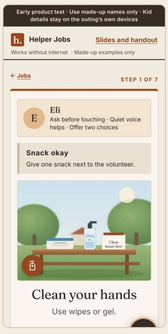
Show less when energy is low

The person picks their energy; the plan shrinks to one step.



Pause, and make it smaller

Save & pause keeps your place; "I'm stuck" shrinks the step.



One picture step, read aloud

A few words, a photo of the real place, and a Help button.

Two apps, built for real life

Both follow the same rules: one step on the screen, big type, a picture, read aloud, and only made-up people in the examples. Swipe to see both.

BY AMISHI MEHTA · FOR SCHOOLWORK

Next Step

For young people who find long instructions, interruptions or explaining their needs hard. Made with neurodivergent students in mind.

What it does: turns a long assignment into one next step, practises words for asking for help, and keeps a note of where to restart. Open Next Step →

FAMILY PROJECT · FOR PRAGNYA OUTINGS

Helper Jobs

Built to give neurodivergent adults a real job at kids' Sunday outings, with a volunteer always close by.

What it does: picture steps read aloud in English, Hindi or Spanish, break reminders, and Help that reaches the lead's phone in about a second. Open Helper Jobs →

The loop on a phone

The phone tour tells the same five chapters, one screen at a time, in the same design as the apps.

Careful claims

The studies suggest these rules; they do not prove the apps work. These are early test versions with made-up people in every example, not tested in a formal study, and they do not diagnose or treat anything.

Closing the loop: do these designs actually help the people they are for? That is where the research goes next. Chapter 5 →

← Chapter 3 · EEG Chapter 5 · What's next → All chapters

← All chapters

CHAPTER 5 · TEST, THEN GO FURTHER

Where this goes next.

The loop does not stop at two studies and two apps. Each turn adds a study, a better design and a write-up others can check. Anything not yet done is labelled as a plan.



The road ahead

Each step builds on the last: data first, then people, then the two together.

NOW · DONE

Two studies, one follow-up, two apps.

The mouse and EEG studies, every result checked and rerunnable, plus experiment 2 on brain-wave bands. Next Step and Helper Jobs working, with the four design rules built in.

Shown: on this site, in the slides and in the phone tour.

NEXT · WITH AN ASDRP MENTOR

Run the three remaining data experiments.

Why the EEG drifts (experiment 4), using the data already here. Then timing (experiment 1, code already written) and groups of brain cells (experiment 3) in the full mouse recordings.

Aim: a first paper, on why the EEG drifts and what brain-wave bands can and cannot fix, for a student research journal such as the *Journal of Emerging Investigators*, plus a poster.

THEN · WITH PEOPLE

Study the design with the people it is for.

Co-design sessions with Pragnya volunteers and families, then a usability test of the four design rules. Consent first, and tasks written down before anyone takes part.

Aim: a second paper, on the design rules, and better apps.

LONGER TERM · THE JOINED-UP QUESTION

Does designing for state help people learn?

Compare the energy-aware design with a fixed one, for the same person over several weeks. Then ask whether the brain findings predict who it helps most.

Aim: one research direction, where neuroscience and human–computer interaction meet.

The next five experiments

One for each link in the loop, with the same steps as chapters 2 and 3.

1 · STARTED · TESTS CHAPTER 2

Does the timing matter?

Does the mouse's state at that exact moment help more than its state a little earlier or later? The code is written in Amishi's repository; it needs the full mouse recordings to run.

2 · DONE · TESTS CHAPTER 3

Brain-wave bands

Bands did better on average (0.566 against 0.481) but two of five stretches still fell just below a coin flip, so by the rule set beforehand it did not hold up. See chapter 3.

3 · PLANNED · TESTS CHAPTER 2

Which brain cells change?

Does state change every brain cell a little, or a few groups a lot? A first look at the first mouse found no clear pattern by where the cells sit. Groups are chosen before looking. Needs the full mouse recordings.

4 · STARTED · TESTS CHAPTER 3

Why does it drift?

How quickly does the eyes-closed pattern change, and can the model keep up if it keeps learning? The stretch-by-stretch scores and the broken-reading check are done.

5 · PLANNED · CLOSES THE LOOP

Do the apps help?

Back to real life: do the four design rules make tasks easier to finish, for the people the apps are for? Needs consent, a task list written in advance and no personal data beyond what the test needs.

What each turn of the loop produces

Not just results: things other people can read, check and use.

PAPERS

Write-ups others can check

Every study ends in a paper with its code, data links and limits, aimed at student journals and science fairs.

POSTERS AND TALKS

Said out loud

The same slides and charts as this site, for symposiums and science fairs.

OPEN TOOLS

Anyone can rerun it

Every analysis reruns with one command, and the apps stay open source.

BETTER APPS

Back to real life

Each design study changes the apps, and each change is tested again.

A DESIGN GUIDE

Rules others can use

A short, plain guide to designing for brain state, built only from findings that held up.

COMMUNITY

Built with people, not for them

Pragnya volunteers and families help decide what gets built and what gets tested.

The rule we keep: a plan stays labelled as a plan until it has code, saved results, a rerun record, its limits written down and a page like the two studies. No paper, poster or result is listed as done before it is.

← Chapter 4 · Design Back to the start →